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```
close all; clear;
```

Homework #2 Answers

in the third figure, the issue with the step response is that the temperature rises above the $T_{\max}$ value of 50 degrees, which represents something that is physically not possible in this system. This error can lead to further inaccurate tracking which increases instabilities in the system.

```
%this creates an issue because the reference input would be wrong leading
%to the controller to decrease the high temperature input and believes it
%needs to compensate even more than it really does.

% I started off with a propotional controller where  $D(s) = K$  and noted that
% the recurring issue with the new plots was a overshoot in the system,
% which led me to implementing a PI controller as it would be better at
% tracking and predict the overshoot in a better manner. After tweaking the
% gain (K) value, I found it to be somewhere around  $K = 1.15$  that gives us
% the desired results.
```

given values from problem description

```
d = 12;
a0 = 0.02;

% using the 2nd-order Pade approximation
G = RR_pade(d, 2, 2) * RR_tf(1, [1/a0 1]);

K = 1.15;
z = 0.02; %% i introduced this value to make a pole-zero cancellation
pf = 0.1;
D = K * pf * RR_tf([1 z], [1 pf 0]);
P = 1; % Loop prefactor for integral control
% Root Locus plot
figure(1);
RR_rlocus(G);
axis([-0.4 0.3 -0.3 0.3]);

% Figure 2: graphing the y(t)
figure(2);
g.T = 200;
RR_step(35 + 10*P*D/(1 + G*D), g);
axis([0 200 32 55]);

% Figure 3: u(t)
figure(3);
RR_step(35 + 10*P*D/(1 + G*D), g);
axis([0 200 35 52]);
```

```
% Figure 4: High-Order Verification 16th-order Pade model
```

```
G_high = RR_pade(d, 16, 13) * RR_tf(1, [1/a0 1]);
```

```
figure(4);
```

```
RR_step(35 + 10*P*G_high*D/(1 + G_high*D), g);
```

```
axis([0 200 32 55]);
```



